Module 3

Transportation Problem – formulation and solution, degeneracy, Assignment Problem – formulation and solution, unbalanced problems, comparison with TP models.

Introduction

- Transportation Problem (TP) and Assignment Problem (AP) are special types of Linear Programming Problems (LPP). They deal with **optimal allocation of resources** in order to **minimize cost or maximize profit**. These models are widely used in logistics, supply chain management, production planning, job allocation, and operations management.

Transportation Problem (TP)

- The Transportation Problem deals with transporting a **single homogeneous commodity** from several **sources (factories/warehouses)** to several **destinations (markets/customers)** at the **minimum transportation cost**, satisfying supply and demand constraints.

Basic Elements of TP

- **Sources (m)**: Supply points (S1, S2, …, Sm)
- **Destinations (n)**: Demand points (D1, D2, …, Dn)
- **Supply (ai)**: Quantity available at source i
- **Demand (bj)**: Quantity required at destination j
- **Cost (cij)**: Cost of transporting one unit from source i to destination j
- **Decision Variable (xij)**: Quantity transported from source i to destination j

Mathematical Formulation of TP

- **Objective Function (Minimization):** Minimize $Z = \Sigma \Sigma c_{ij} x_{ij}$
- **Subject to constraints:**
- Supply constraints: $\Sigma x_{ij} = a_i$ for each source i
- Demand constraints: $\Sigma x_{ij} = b_j$ for each destination j
- Non-negativity: $x_{ij} \geq 0$

Balanced and Unbalanced Transportation Problems

- **Balanced TP**
- If total supply = total demand, the problem is **balanced**.
- $\Sigma a_i = \Sigma b_j$
- **Unbalanced TP**
- If total supply ≠ total demand, the problem is **unbalanced**.
- **Balancing Method:**
- If supply > demand → Add a **dummy destination** with demand equal to excess supply and zero cost.
- If demand > supply → Add a **dummy source** with supply equal to excess demand and zero cost.

Solution Procedure of TP

- **Step 1: Find an Initial Basic Feasible Solution (IBFS)**
- Common methods:
- North-West Corner Rule
- Least Cost Method
- Vogel’s Approximation Method (VAM – best initial solution)
- **Step 2: Optimality Test**
- MODI Method (u–v method)
- Stepping Stone Method

Example 1: Balanced Transportation Problem (North-West Corner Rule) Problem Statement

A company has **2 factories** and **3 warehouses**. The transportation cost per unit is given below:

	W1	W2	W3	Supply
F1	2	3	1	30
F2	5	4	8	40
Demand	20	25	25	

- **Step 1: Check Balance**
- Total Supply = 30 + 40 = 70 Total Demand = 20 + 25 + 25 = 70
- ✓ Problem is **Balanced**

- **Step 2: Apply North-West Corner Rule**
- Start from the top-left (north-west) cell.
- Allocate min(30,20) = 20 to (F1,W1)
- F1 supply left = 10
- W1 demand satisfied

	W1	W2	W3	Supply
F1	2	3	1	30 (10)
F2	5	4	8	40
DEMAND	20	25	25	

- Move to (F1,W2): Allocate min(10,25) = 10
- F1 supply exhausted
- W2 demand left = 15

	W1	W2	W3	Supply
F1	2	3	1	10
F2	5	4	8	40
DEMAND	20	25 (15)	25	

- Move to (F2,W2): Allocate min(40,15) = 15
- F2 supply left = 25

	W1	W2	W3	Supply
F1	2	3	1	10
F2	5	4	8	40 (25)
DEMAND	20	15	25	

- Move to (F2,W3): Allocate min(25,25) = 25
- All supply and demand satisfied

	W1	W2	W3	Supply
F1	2	3	1	10
F2	5	4	8	25
DEMAND	20	15	25	

- **Step 3: Calculate Total Cost**
- Cost = (20×2) + (10×3) + (15×4) + (25×8)
- Cost = 40 + 30 + 60 + 200 = **₹330**

Route	Units Sent	Cost per unit
F1 → W1	20	2
F1 → W2	10	3
F2 → W2	15	4
F2 → W3	25	8

- **Answer**
- Initial Transportation Cost = **₹330**

Least Cost Method (LCM)

- The Least Cost Method is used to find an Initial Basic Feasible Solution (IBFS) of a Transportation Problem.
- 1. What is Least Cost Method?
- In the Least Cost Method, we:
- Look for the cell with the minimum transportation cost
- Allocate as much as possible to that cell
- Adjust supply and demand
- Repeat the process until all supply and demand are satisfied
- 👉Idea: Send goods through the cheapest routes first.

- 2. Steps of Least Cost Method (Very Important for Exams)
- Identify the lowest cost cell in the entire table
- Allocate the minimum of supply and demand to that cell
- Reduce the corresponding supply or demand
- Cross out the row or column that becomes zero
- Repeat until all requirements are met

3. Numerical Example (Step-by-Step)

Problem Statement

	D1	D2	D3	Supply
S1	6	8	10	20
S2	7	11	11	30
Demand	10	15	25	

- Step 1: Check Balance
- Total Supply = 20 + 30 = 50 Total Demand = 10 + 15 + 25 = 50
- ✓ Balanced Transportation Problem
- Step 2: Find the Least Cost Cell
- The smallest cost in the table is 6 at (S1, D1)

- Step 3: First Allocation
- Allocate min(20, 10) = 10 units to (S1, D1)
- Updated:
- S1 supply left = 10
- D1 demand satisfied → column D1 crossed out

Step 4: Next Least Cost Cell

- Remaining costs:

	D2	D3	Supply
S1	8	10	10
S2	11	11	30
Demand	15	25	

Lowest cost = 8 at (S1, D2)
Allocate min(10, 15) = 10 units
Updated:
•S1 supply exhausted → row S1 crossed out
•D2 demand left = 5

- Step 5: Continue Allocation
- Only S2 is left:
- Allocate 5 units to (S2, D2)
- Allocate 25 units to (S2, D3)

Final Allocation Table

	D1	D2	D3	Supply
S1	10	10	–	20
S2	–	5	25	30
Demand	10	15	25	

4. Total Transportation Cost Calculation

Multiply allocation × cost for each filled cell:

- (10 × 6) = 60
- (10 × 8) = 80
- (5 × 11) = 55
- (25 × 11) = 275

- Total Cost:
- 60 + 80 + 55 + 275 = ₹470

- 5. Final Answer

- Total transportation cost using Least Cost Method = **₹470**

Vogel’s Approximation Method (VAM)

- Vogel’s Approximation Method is an improved technique to obtain a good Initial Basic Feasible Solution (IBFS) for a Transportation Problem. It usually gives a solution very close to the optimal solution.
- 1. What is Vogel’s Approximation Method?
- VAM considers not only the lowest cost but also the penalty for not choosing the lowest-cost route.
- 👉Penalty = loss incurred if the cheapest route is not selected.

- 2. What is Penalty? (Very Important ⭐)
- Definition:
- Penalty is the difference between the smallest and the next smallest cost in a row or column.
- Why penalty?
- It shows how important it is to allocate in that row or column
- Higher penalty → greater loss if we don’t choose the cheapest cell

3. Steps of Vogel's Approximation Method

- For each row and column, find the penalty
- Select the row or column with the highest penalty
- In that row/column, choose the cell with the lowest cost
- Allocate the minimum of supply and demand
- Adjust supply and demand, strike out satisfied row/column
- Repeat until all allocations are done

4. Numerical Example (Step-by-Step)

- Problem Statement

	D1	D2	D3	Supply
S1	2	3	1	20
S2	5	4	8	30
S3	5	6	8	25
Demand	15	25	35	

Step 1: Check Balance

Total Supply = 20 + 30 + 25 = 75

Total Demand = 15 + 25 + 35 = 75

✓ Balanced Problem

- Step 2: Calculate Row Penalties
- S1 costs: 1, 2, 3 → Penalty = 2 − 1 = 1
- S2 costs: 4, 5, 8 → Penalty = 5 − 4 = 1
- S3 costs: 5, 6, 8 → Penalty = 6 − 5 = 1

- Step 3: Calculate Column Penalties

- D1 costs: 2, 5, 5 → Penalty = 5 – 2 = 3
- D2 costs: 3, 4, 6 → Penalty = 4 – 3 = 1
- D3 costs: 1, 8, 8 → Penalty = 8 – 1 = 7

- **Step 4: Choose Highest Penalty**
- Highest penalty = 7 (Column D3)
- Choose the lowest cost in column D3 → 1 at (S1, D3)
- **Step 5: Allocate**
- Allocate min(20, 35) = 20 to (S1, D3)
- Updated:
- S1 exhausted → row S1 removed
- D3 demand left = 15

Step 6: Recalculate Penalties (Reduced Table)

	D1	D2	D3	Supply
S2	5	4	8	30
S3	5	6	8	25
Demand	15	25	15	

- Penalties:
- Rows S2 & S3 → Penalty = 1
- Column D1 → Penalty = 0
- Column D2 → Penalty = 2
- Column D3 → Penalty = 0

- Step 7: Next Allocation
- Highest penalty = 2 (Column D2)
- Lowest cost in D2 = 4 at (S2, D2)
- Allocate min(30, 25) = 25
- Updated:
- S2 supply left = 5
- D2 satisfied → column removed

- Step 8: Final Allocations
- Remaining:

	D1	D3	Supply
S2	5	8	5
S3	5	8	25
Demand	15	15	

Allocate:

(S2, D1) = 5

(S3, D1) = 10

(S3, D3) = 15

5. Final Allocation Table

Route	Allocation	Cost
S1 → D3	20	1
S2 → D2	25	4
S2 → D1	5	5
S3 → D1	10	5
S3 → D3	15	8

- 6. Total Transportation Cost
- = (20×1) + (25×4) + (5×5) + (10×5) + (15×8)
- = 20 + 100 + 25 + 50 + 120
- Total Cost = ₹315

- 7. Why VAM is Better ⭐
- Considers cost difference (penalty)
- Avoids poor early allocations
- Gives near-optimal solution
- Vogel’s Approximation Method is an IBFS technique that uses penalties to decide the order of allocations in a transportation problem.

- NWCR → No thinking about cost
- LCM → Think about cheapest cell
- VAM → Think about loss if cheapest is missed

What is Degeneracy? (Transportation Problem)

- A Transportation Problem is said to be degenerate when the number of allocations is less than

- $(m + n \ -1)$

where

- m = number of rows (sources / factories)
- n = number of columns (destinations / warehouses)
- Why (m + n − 1)?

- In any valid transportation solution:

- We must have exactly (m + n − 1) occupied cells (basic variables)

- Otherwise, further calculations (like MODI method) will not work properly

- Example Table: 2 × 3 Transportation Problem
- m = 2 (rows)
- n = 3 (columns)
- Required allocations:

- $m + n - 1 = 2 + 3 - 1 = 4$

	W1	W2	W3
F1	20	10	–
F2	–	15	–

Number of allocations = 3
But we need 4 allocations
❌ This is a degenerate solution

Why Does Degeneracy Occur?
Degeneracy usually happens when:
Supply and demand become zero at the same time
A row and a column are crossed out together
Careless allocation in IBFS methods

How to Handle Degeneracy?

- Solution:
- 👉Add a very small quantity ε (epsilon) to an empty cell
- ε is almost zero
- It does not change total cost
- It is used only to maintain (m + n – 1) allocations

- Degeneracy is handled by introducing a very small quantity ε in an unoccupied cell.
- If allocations < $(m + n - 1)$
- → Degeneracy
- → Add ε

Comparison: Transportation Problem vs Assignment Problem

Feature	Transportation Problem (TP)	Assignment Problem (AP)
Objective	Minimize transportation cost	Minimize assignment cost
Supply	Any positive value	1
Demand	Any positive value	1
Number of allocations	(m+n-1)	(n)
Splitting allowed	Yes	No
Solution methods	NWCM, LCM, VAM	Hungarian Method
Nature	General	Special case of TP

- Assignment Problem: A special transportation problem with unit supply and unit demand.

- Unbalanced Assignment Problem: When number of jobs and persons are unequal.

- Solution Method: Hungarian Method.

- Assignment Problem:
- One person → One job
- One job → One person
- Supply = Demand = 1
- Solved by Hungarian Method

- “Transportation decides how much to send,
- Assignment decides who does what.”